

IC 231: Measurement and Instrumentation Practicum

Course Code	IC 231
Course Name	Measurement and Instrumentation Practicum
L-T-P-C	2-0-2-3
Prerequisites	IC 152 Data Science I/Computer and Data Science, IC 161 Applied Electronics, IC 161P Applied Electronics Practicum
Intended for	All the B.Tech branches
Mutual Exclusion	None
Approval	24th Senate (Feb 2020), in the B.Tech EE curriculum; 44th BoA (Nov 2021)
Revised	34th Senate (Mar 2022) - credits changed to 2-0-2-3

Preamble

The objective of the course is to provide an overview of measurement and instrumentation techniques used in engineering disciplines viz. mechanical, chemical, electrical, electronics, etc.

Upon completion of this course, the students will be able to

1. Identify and differentiate between different classes of sensors and actuators and choose the right component for application-specific purposes.
2. Use commonly used open-source microcontroller and microcomputer platforms for signal generation, sensor data acquisition, device actuation and automation.
3. Analyze and compare design approaches and architectures of real-world measurement systems, and identify key parameters for assessing their performance.
4. Design open and closed loop actuation, automation and measurement systems for their own discipline specific applications.

Course Modules

- **Topic 1: Measurement fundamentals:** Fundamental and derived quantities: static and dynamic, understanding, sensitivity, stability, resolution, accuracy, precision, calibration, and types of errors. (2 Hours)
- **Topic 2: Microcontroller and microcomputer-based data acquisition and automation:** Introduction to microcontroller and microcomputer (e.g., Arduino, Raspberry Pi), interfacing considerations (e.g., communication protocols, use of multiplexers), device control and data acquisition using Python, simple routines for signal processing and analysis examples. (4 Hours)
- **Topic 3: Principles of Instrumentation:** Sensor interrogation principles - e.g., using bridge circuits, signal amplification, signal conditioning (transduction, linearization), phase measurements, active and passive filters, isolation and shielding, elements of control theory, digital data acquisition principles using ADC/DACs. (6 Hours)
- **Topic 4: Sensors and Actuators:** Sensor classification, static and dynamic characteristics, Sensor examples from different domains – mechanics (e.g., strain gauge, accelerometer, LVDT), thermodynamics (e.g., thermistors, thermocouples), fluidics (e.g., venturimeter, ultrasonic flowmeter), biomedical (e.g., electrodes), electromagnetics (e.g., Hall sensor). Actuator examples – piezo-electric transducer, stepper motor. (10 Hours)

- **Topic 5: Measurement System Examples:** Systems approach to design, Noise and SNR considerations (e.g. application of Friis equation), Analysis of design of real-world measurement systems – for e.g. structural health monitoring, biomedical systems (e.g. ECG, EMG, EEG), air-quality monitoring using electrochemical sensors, LIDAR, contact-based (e.g. LVDT) and non-contact (e.g. ultrasonic, optical) distance measurement systems (6 Hours)

Laboratory Modules (28 hours)

1. Stepper motor controller
2. Temperature measurement using thermal sensors,
3. Flow measurement,
4. Experiment on LVDT,
5. Level/distance measurement using contact-less sensor,
6. Vibration/Sound measurement and FFT based analysis,
7. Chemical composition detection,
8. Bio-signal measurement,
9. Project.

Text Books

1. Fraden, Jacob. *Handbook of modern sensors*. Springer Science+Business Media, 2010.
2. Khandpur, R., *Handbook of Biomedical Instrumentation*. 3/e, Tata McGraw Hill, 2014.

References

1. Doebelin, E. O., Manik, D. N., *Measurement Systems*, 6/e, Tata McGraw Hill India, 2011.
2. Singh, S. K., *Industrial Measurement and Control*, 2/e, Tata McGraw Hill India, 2003.
3. Webb, A. G., *Principles of Biomedical Instrumentation*, Cambridge University Press UK, 2018.